Midterm Project

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Outline

1. **Supplementary Difference Sets(SDS)**
   Definition and constraints.
2. **Program Steps**
   Subgroup selection, orbit generation.
3. **Flow Chart**
   Search flow for (X, Y) using orbit structure.
4. **Construction of PCPs**
5. **Results**
6. **Summary and Reflections**
7. **References**

Why Introduce SDS?

- PCP is defined via periodic autocorrelation:

$$\rho(a;u) = \sum_{i=0}^{L-1} a_i\, a_{i+u} \pmod{L}.$$

- Autocorrelation implicitly counts "how many positions match" under a shift.
- SDS makes this counting *explicit* by working with sets and overlaps.

Key strategy

1. Convert ± 1 sequences a, b into sets X, Y of indices where -1 occurs.
2. Define a counting function $N_X(u)$ that measures the overlap of X with its shifted version.
3. Express $\rho(a;u)$ in terms of $N_X(u)$.
4. Show that the PCP condition is equivalent to an SDS condition on $N_X(u)$ and $N_Y(u)$.

From Sequences to Sets: Meaning of X and Y

Consider two ± 1 sequences

$$a = (a_0, a_1, \ldots, a_{L-1}), \qquad b = (b_0, b_1, \ldots, b_{L-1}).$$

Definition of X and Y:

$$X = \{\, i : a_i = -1 \,\}, \qquad Y = \{\, i : b_i = -1 \,\}.$$

- X collects all positions where $a_i = -1$.
- Y collects all positions where $b_i = -1$.
- Let $r = |X|$ and $s = |Y|$.

Sequence vs. set viewpoint

- Sequence: $a_i \in \{\pm 1\}$, $b_i \in \{\pm 1\}$.
- Set: indices of -1 are stored in X and Y.
- We can also write $a_i = 1 - 2x_i$ where $x_i = 1$ if $i \in X$ and 0 otherwise.

Definition of $N_X(u)$

Shifted set:

$$X + u = \{\, i + u \pmod{v} : i \in X \,\}.$$

Overlap counting function:

$$N_X(u) = \left|\{\, i : i \in X,\ i + u \pmod{v} \in X \,\}\right| = |\, X \cap (X + u)\,|.$$

- $N_X(u)$ counts how many elements of X remain inside X after a shift by u.
- Similarly, $N_Y(u) = |\, Y \cap (Y + u)\,|$.
- These overlap counts will later replace autocorrelation values.

Intuition

If $N_X(u)$ is large, then X and $X + u$ have a strong overlap at shift u.

Definition of SDS

Let $X, Y \subset \mathbb{Z}_v$ with $|X| = r$ and $|Y| = s$. Define the overlap counts

$$N_X(u) = |X \cap (X + u)|, \qquad N_Y(u) = |Y \cap (Y + u)|.$$

Supplementary Difference Set (SDS)

The pair (X, Y) is called an SDS with parameters $(v; r, s; \lambda)$ if

$$N_X(u) + N_Y(u) = \begin{cases} r + s, & u = 0, \\ \lambda, & u \neq 0. \end{cases}$$

- For $u = 0$, the overlap counts are simply $N_X(0) = r$ and $N_Y(0) = s$.
- For all nonzero shifts u, the total overlap $N_X(u) + N_Y(u)$ is a constant λ.
- This uniformity condition is the set-theoretic counterpart of the PCP condition.

Deriving λ (Step 1: Autocorrelation)

- Let $a_i = 1 - 2x_i$ with $x_i = \mathbf{1}_{i \in X}$. Then

$$\rho(a; u) = \sum_{i=0}^{v-1} a_i \, a_{i+u(\bmod v)} = \sum_{i=0}^{v-1} (1 - 2x_i)(1 - 2x_{i+u(\bmod v)}).$$

- Expanding the product:

$$\rho(a; u) = \sum_{i=0}^{v-1} \left[1 - 2x_i - 2x_{i+u(\bmod v)} + 4x_i x_{i+u(\bmod v)}\right].$$

- Using

$$\sum_{i=0}^{v-1} x_i = r = |X|, \qquad \sum_{i=0}^{v-1} x_i x_{i+u(\bmod v)} = N_X(u),$$

we obtain

$$\rho(a; u) = v - 4r + 4N_X(u).$$

Deriving λ (Step 2: Using PCP condition)

- From Step 1:
$$\rho(a;u) = v - 4r + 4N_X(u), \qquad \rho(b;u) = v - 4s + 4N_Y(u).$$
- Add the two expressions:
$$\rho(a;u) + \rho(b;u) = 2v - 4(r+s) + 4\big(N_X(u) + N_Y(u)\big).$$
- PCP requires (for $u \neq 0$):
$$\rho(a;u) + \rho(b;u) = 0.$$
- Therefore
$$0 = 2v - 4(r+s) + 4\big(N_X(u) + N_Y(u)\big),$$
and solving gives
$$N_X(u) + N_Y(u) = r + s - \frac{v}{2} \triangleq \lambda.$$

$$\boxed{\lambda = r + s - \frac{v}{2}}$$

PCP $\Rightarrow$ SDS: A Successful Example

- Consider the complementary pair:

$$a = [1, 1, 1, -1], \qquad b = [1, 1, -1, 1].$$

Step 1. Convert to sets.

$$X = \{3\}, \qquad r = 1, \qquad Y = \{2\}, \qquad s = 1.$$

Step 2. Compute λ.

$$\lambda = r + s - \frac{v}{2} = 1 + 1 - 2 = 0.$$

Goal: show that $N_X(u) + N_Y(u) = 0$ for all $u \neq 0$.

PCP $\Rightarrow$ SDS: Overlap Counts

- $N_X(u)$ with $X = \{3\}$:

u	$N_X(u)$
0	1
1	0
2	0
3	0

- $N_Y(u)$ with $Y = \{2\}$:

u	$N_Y(u)$
0	1
1	0
2	0
3	0

- Combined:

$$N_X(u) + N_Y(u) = \begin{cases} 2, & u = 0, \\ 0, & u \neq 0. \end{cases}$$

From Sequence Sums A, B to (v, r, s, λ)

- We look for two binary sequences $a, b \in \{\pm 1\}^v$.
  Let $A = \sum_{i=0}^{v-1} a_i,\ B = \sum_{i=0}^{v-1} b_i$.
- For a ± 1 sequence, if r entries are -1, then
  $A = (v - r) - r = v - 2r \Rightarrow r = \frac{v-A}{2}$. Similarly, $s = \frac{v-B}{2}$.
- In our case:
  $$v = 74, \quad A = 2, \quad B = 12$$
  giving
  $$r = \frac{74-2}{2} = 36, \qquad s = \frac{74-12}{2} = 31.$$
- From the SDS–PCP relation:
  $$\lambda = r + s - \frac{v}{2} = 36 + 31 - 37 = 30.$$

Thus $(v, r, s, \lambda) = (74, 36, 31, 30)$**.**

Step 1: Choosing the Subgroup H

- From prime factorization, we have $74 = 2 \times 37$, so

$$|\mathbb{Z}_{74}^*| = \varphi(74) = 74\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{37}\right) = 36$$

  Therefore, the possible subgroup sizes must divide 36:

$$|H| \in \{1,\ 2,\ 3,\ 4,\ 6,\ 9,\ 12,\ 18,\ 36\}$$

- For $g \in \mathbb{Z}_{74}^*$, the order of g is the smallest m such that

$$g^m \equiv 1 \pmod{74}.$$

  The subgroup generated by g is

$$\langle g \rangle = \{g^0,\ g^1,\ g^2,\ \ldots,\ g^{m-1}\}$$

- For $L = 74$, we choose

$$H = \{1,\ 47,\ 63\}.$$

  Check : $47^1 \equiv 47, \qquad 47^2 \equiv 63, \qquad 47^3 \equiv 1 \pmod{74}$

Step 2 : Determining $|J|$ and $|K|$

- To determine the parameters (r, s, λ) of the SDS, solve :

$$\begin{cases} r(r-1) + s(s-1) = \lambda(v-1) \\ \lambda = r + s - \dfrac{v}{2} \end{cases} \qquad (v = 74)$$

  We choose $r = 36, s = 31, \lambda = 30$

- Since $|H| = 3$, every nontrivial H-orbit has size 3
  For X, we use only J nontrivial 3-orbits:

$$|X| = r = 36 = 3|J| \quad \Rightarrow \quad |J| = 12$$

  For Y, we use K nontrivial 3-orbits plus one trivial orbit:

$$|Y| = s = 31 = 3|K| + 1 \quad \Rightarrow \quad |K| = 10$$

Step 3: H-Orbits in $\mathbb{Z}_{74}$ (1)

Recall the subgroup $H = \{1, 47, 63\}, \qquad |H| = 3$

Definition of the H-orbit of j : $(j = 0, 1, \ldots, 73)$

$$O_j \;=\; \{\, h \cdot j \pmod{74} \mid h \in H \,\}.$$

Trivial orbits:

- $j = 0$:
  $h \cdot 0 \equiv 0 \pmod{74},\ \forall h \in H \;\Rightarrow\; O_0 = \{0\}.$
- $j = 37$:
  $1 \cdot 37 = 37,\ 47 \cdot 37 \equiv 37,\ 63 \cdot 37 \equiv 37 \;\Rightarrow\; O_{37} = \{37\}.$

Step 3: H-Orbits in $\mathbb{Z}_{74}$ (2)

Nontrivial orbits:

for every $j \notin \{0, 37\}$, the orbit O_j has 3 distinct elements, so the number of nontrivial orbits is $\dfrac{74-2}{3} = 24$.

All orbit representatives under the action of H:

```
   reps: [1 2 3 4 5 6 7 9 10 11 12 14 15 17 18 21 22 23 28 29 34 36 42 55]
 orbits: {1×24 cell}
trivial: [0 37]
```

Example:

$j = 2: \ 1 \cdot 2 = 2, \ 47 \cdot 2 \equiv 20, \ 63 \cdot 2 \equiv 52 \ \Rightarrow \ O_2 = \{2, 20, 52\}.$

$j = 19: \ 1 \cdot 19 = 19, \ 47 \cdot 19 \equiv 5, \ 63 \cdot 19 \equiv 13 \ \Rightarrow \ O_{19} = \{19, 5, 13\}.$

the smallest element is 5, so 19 is **not** an orbit representative.

Step 4: From J, K to Sets X, Y and Sequences a, b

- Given a choice of J and K, we build the sets X and Y:

$$X = \bigcup_{j \in J} O_j, \qquad Y = \left(\bigcup_{k \in K} O_k \right) \cup \{37\}$$

- Then checks whether

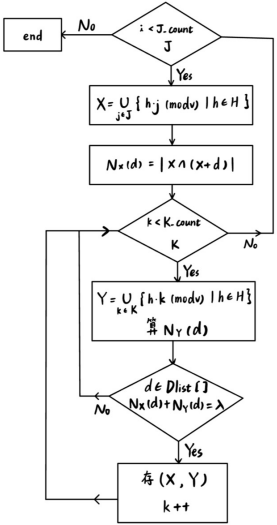
$$N_X(d) + N_Y(d) = \lambda$$

- Convert X to the ± 1 sequence A:

$$a_i = \begin{cases} -1, & i \in X, \\ +1, & i \notin X. \end{cases}$$

- Convert Y to the sequence B in the same way

Flow chart


end
No
i < J.count
J
Yes
X = ∪ {h·j (mod v) | h ∈ H}
j∈J
Nx(d) = |X ∩ (X+d)|
k < K.count
K
No
Yes
Y = ∪ {h·k (mod v) | h ∈ H}
k∈K
并 NY(d)
d ∈ Dlist[]
Nx(d)+NY(d) = λ
No
Yes
存 (X, Y)
k++

Construction of PCPs (1/4)

- **Goal:** Find a longer periodic complementary pair (PCP).
- **Known:** We already have a PCP (A, B) of length 82.
- **Strategy:**
  - From [1], we know that a PCP of length 164 can be obtained from one of length 82 via the *multiplying* construction*.
  - In [2], an explicit construction method for length 164 is given†.

*[1]D. Z. Djokovic and I. S. Kotsireas, "Some new periodic golay pairs," *Numerical Algorithms*, vol. 69, no. 3, pp. 523–530, Sep. 2015.

†[2]L. Bomer and M. Antweiler, "Periodic complementary binary sequences," *IEEE Transactions on Information Theory*, vol. 36, no. 6, pp. 1487–1494, 1990.

Construction of PCPs (2/4)

Let (A, B) be a PCP of length $N = 82$:

$$A = (a_0, a_1, \ldots, a_{81}), \qquad B = (b_0, b_1, \ldots, b_{81}).$$

We also use the reversed and reversed–negated versions:

$$\tilde{B} = (b_{81}, b_{80}, \ldots, b_1, b_0), \qquad -\tilde{A} = (-a_{81}, -a_{80}, \ldots, -a_1, -a_0).$$

Construction of PCPs (3/4)

Define two sequences E and F of length 164 by interleaving.

Even / odd positions: for $0 \le k < 82$,

$$E(2k) = a_k, \qquad E(2k+1) = b_{81-k},$$

$$F(2k) = b_k, \qquad F(2k+1) = -a_{81-k}.$$

Equivalently, E and F are obtained by

$$E = \text{interleave}\big(A,\ \tilde{B}\big), \qquad F = \text{interleave}\big(B,\ -\tilde{A}\big).$$

Construction of PCPs (4/4)

Even / odd positions can be separated:

$$E(2k) = a_k, \quad E(2k+1) = b_{81-k},$$
$$F(2k) = b_k, \quad F(2k+1) = -a_{81-k},$$

for $0 \leq k < 82$.

Equivalent sequence form:

$$\begin{aligned} E &= (e_0, e_1, \ldots, e_{163}) \\ &= (a_0, b_{81}, a_1, b_{80}, \ldots, a_{81}, b_0), \end{aligned}$$

$$\begin{aligned} F &= (f_0, f_1, \ldots, f_{163}) \\ &= (b_0, -a_{81}, b_1, -a_{80}, \ldots, b_{81}, -a_0). \end{aligned}$$

Results of L=74 & PAPR

Metric	Value
Total PCP Pairs	276
Minimum PAPR	2.779042(4.4390dB)
Average PAPR	4.043040(6.0671dB)

$$
\begin{aligned}
A = (&1, -1, -1, 1, 1, 1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, -1, -1, 1, \\
&-1, -1, 1, -1, -1, -1, -1, 1, -1, -1, 1, -1, -1, -1, 1, -1, 1, 1, -1, 1, \\
&1, 1, 1, -1, 1, -1, 1, -1, 1, -1, 1, -1, -1, -1, 1, 1, 1, 1, -1, -1, \\
&-1, 1, -1, -1, 1, -1, 1, 1, 1, 1, 1, -1, 1, 1)
\end{aligned}
$$

$$
\begin{aligned}
B = (&1, -1, 1, -1, 1, -1, 1, 1, 1, -1, 1, 1, -1, -1, 1, 1, -1, 1, 1, -1, \\
&1, 1, -1, 1, 1, 1, 1, 1, -1, -1, 1, -1, 1, 1, 1, 1, -1, -1, 1, 1, \\
&1, -1, -1, 1, 1, 1, -1, -1, -1, -1, -1, -1, 1, -1, -1, 1, -1, 1, -1, 1, \\
&1, 1, -1, -1, -1, 1, 1, -1, 1, 1, 1, 1, -1, 1)
\end{aligned}
$$

Results of L=82 & PAPR

Metric	Value
Total PCP Pairs	36
Minimum PAPR	3.024673 (4.8068dB)
Average PAPR	3.866760 (5.8735dB)

$$
\begin{aligned}
A = (&1, -1, -1, -1, -1, 1, -1, -1, -1, 1, 1, -1, 1, -1, -1, 1, 1, 1, 1, 1, \\
&-1, 1, 1, -1, 1, -1, -1, 1, 1, -1, 1, -1, -1, -1, 1, 1, -1, -1, 1, 1, \\
&-1, 1, 1, -1, 1, -1, -1, 1, 1, 1, -1, -1, 1, -1, 1, 1, 1, -1, -1, -1, \\
&-1, -1, -1, 1, -1, 1, -1, 1, 1, -1, 1, -1, -1, -1, -1, -1, 1, -1, 1, -1, \\
&-1, -1)
\end{aligned}
$$

$$
\begin{aligned}
B = (&-1, -1, 1, -1, -1, -1, 1, -1, 1, -1, 1, -1, -1, -1, 1, 1, 1, 1, 1, 1, \\
&1, -1, 1, 1, 1, 1, 1, 1, -1, -1, -1, 1, 1, 1, -1, 1, 1, -1, -1, -1, \\
&-1, 1, 1, 1, 1, -1, 1, 1, 1, -1, -1, 1, -1, -1, -1, -1, 1, 1, -1, 1, -1, \\
&1, 1, 1, 1, -1, 1, -1, 1, 1, -1, -1, -1, -1, 1, 1, -1, 1, 1, 1, -1, \\
&1, 1)
\end{aligned}
$$

Results of L=122 and L=164 & PAPR

Metric	L = 122	L = 164
Total PCP Pairs	6	36
Minimum PAPR	3.671929(5.6789dB)	3.082117(4.8885dB)
Average PAPR	4.595595(6.6234dB)	4.275740(6.3101dB)

Summary and Reflections

- **Execution Speed Optimization**
  Full search for sequences of length $L = 34$ completed within 30 seconds.
- **Acquiring Additional Hardware Resources**
- **Implementation of the Paper Method**

Thank you !!

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References I

[1] D. Z. Djokovic and I. S. Kotsireas, “Some new periodic golay pairs,” *Numerical Algorithms*, vol. 69, no. 3, pp. 523–530, Sep. 2015.

[2] L. Bomer and M. Antweiler, “Periodic complementary binary sequences,” *IEEE Transactions on Information Theory*, vol. 36, no. 6, pp. 1487–1494, 1990.

[3] D. Z. Djokovic and I. S. Kotsireas, “Compression of periodic complementary sequences and applications,” *Designs, Codes and Cryptography*, vol. 74, no. 2, pp. 365–377, Feb. 2015.

[4] D. Z. Djokovic and I. S. Kotsireas, “New results on d-optimal matrices,” *arXiv preprint arXiv:1103.3626*, 2011.

All Results

Metric	L = 74	L = 82	L = 122	L = 164
Total Pairs	276	36	6	36
Min PAPR	2.779042(4.4390dB)	3.024673 (4.8068dB)	3.671929(5.6789dB)	3.082117(4.8885dB)
Avg. PAPR	4.043040(6.0671dB)	3.866760 (5.8735dB)	4.595595(6.6234dB)	4.275740(6.3101dB)